# Pell Centers and Fused Bounds

Explicit Diophantine presentations of Con(ZFC)

Prepared for Vladimir Reshetnikov and the ProveIt research program
Research and artifact development with AI assistance

25 October 2026

**Abstract**

We give three arithmetic reductions of explicit consistency equations in the pinned ProveIt snapshot. A polynomial Pell center replaces a large constant in the growth certificate and yields a quartic in 212 unrestricted integer variables. Its small-coefficient version has 152 quadratic residual squares, 21,556 collected monomials, and coefficient height 6,432; a sparse version has 5,290 monomials and height 277,420. A certificate shared between the Pell-index cap and the binomial-remainder bound removes three integer variables. Combining it with a linear Pell-orientation numerator gives an 82-variable equation of exact degree 16,964 with 1,723 live binary arithmetic operations; companions have 81 variables at degree 2,097,732 and 106 variables at degree 32. Separately, an odd-index Pell residue eliminates a native transport quotient, giving a 13-positive-variable equation of degree 174 with a 69-operation evaluator.

Complete new ordinary proofs, literal equations, finite algebraic certificates, independent checkers, and reproducible construction programs accompany this report. Composed with the explicitly identified inherited logical calibrations, the proofs are formalizable in ZFC and establish that the respective equations have a zero exactly when ZFC has a finite refutation. Global minimality and a fresh end-to-end proof-assistant compilation remain open.

## Contents

1 Results and their exact scope 2

2 The arithmetic and logical dependency boundary 3

3 A polynomial Pell center with small coefficients 3

4 One certificate across two arithmetic stages 6

5 Two Pell roots carry two quotients 9

6 Reproduction, evidence, and integration 13

7 Further research questions 14

A The complete small-coefficient quartic 16

# 1 Results and their exact scope

Throughout, Con(ZFC) means absence of a finite syntactic refutation in the fixed effective ZFC presentation used by ProveIt. Effective translations between the standard presentations give the usual consistency statement. All polynomial coefficients are integers. A named register in a circuit is an abbreviation for a polynomial, never an additional quantified variable.

**Theorem 1.1** (Consistency equations)**.** *For each integer-domain polynomial $P$ in Table 1, with its indicated number $n$ of inputs, the ordinary proofs in this report, composed with the inherited calibration identified in Section 2, establish*

$$\mathsf{ZFC} \vdash \big(\neg\exists \boldsymbol{x} \in \mathbb{Z}^n \; P(\boldsymbol{x}) = 0\big) \longleftrightarrow \mathrm{Con}(\mathsf{ZFC}).$$

*For the positive-domain polynomials, replace $\mathbb{Z}^n$ by $\mathbb{Z}_{>0}^{13}$. Their explicit 39-integer lifts have the same nonsolvability equivalence over $\mathbb{Z}^{39}$.*

| Polynomial | Domain | Inputs | Exact degree | Further size data |
|---|---|---|---|---|
| $P_{212}^{\mathrm{small}}$ | $\mathbb{Z}$ | 212 | 4 | 152 squares; height 6,432 |
| $P_{212}^{\mathrm{sparse}}$ | $\mathbb{Z}$ | 212 | 4 | 5,290 monomials; height 277,420 |
| $P_{106}$ | $\mathbb{Z}$ | 106 | 32 | 47 squares; 1,830 operations |
| $P_{82}$ | $\mathbb{Z}$ | 82 | 16,964 | 23 squares; 1,723 operations |
| $P_{81}$ | $\mathbb{Z}$ | 81 | 2,097,732 | 22 squares; 1,711 operations |
| $C_{13}$ | $\mathbb{Z}_{>0}$ | 13 | 174 | 69 operations |
| $C_{13}^{\mathrm{oriented}}$ | $\mathbb{Z}_{>0}$ | 13 | 170 | 70 operations |
| $\widetilde{C}_{39}$ | $\mathbb{Z}$ | 39 | 348 | 160 operations |
| $\widetilde{C}_{39}^{\mathrm{oriented}}$ | $\mathbb{Z}$ | 39 | 340 | 161 operations |

Table 1: Explicit equations supplied in this submission. Operations count live binary $+, -, \times$ gates with integer leaves.

The principal quartic is literally

$$P_{212}^{\mathrm{small}}(x_0, \ldots, x_{211}) = \sum_{j=0}^{151} R_j(x_0, \ldots, x_{211})^2 = 0, \qquad (x_0, \ldots, x_{211}) \in \mathbb{Z}^{212}. \tag{1}$$

Every $R_j$ is printed in Appendix A, and the JSON file `quartic212/reduced/equation.json` gives the same 152 literal quadratic polynomials. The full collected quartic is also supplied. Thus the definition contains no implicit universal polynomial, unspecified search procedure, or variable exponent.

The high-degree direct equations are supplied as acyclic arithmetic graphs, with independent input lists and fixed residual outputs. Their specification is just as finite and literal, while avoiding expansion of very large polynomials. The 13-positive-variable formula is short enough to print in full in the main text.

## 1.1 What improves, and what remains a tradeoff

The inspected repository revision is `b42c796d1f5de9912a576bc2fd77b778ae7402bf`. It includes the latest incoming archives, not only the current root display. The comparisons below refer to actual domain and arithmetic data.

| Matched direct family | Integer inputs | Degree | Operations | Squares |
|---|---|---|---|---|
| Incoming projected source | 109 | 32 | 1,895 | 48 |
| New fused and oriented | 106 | 32 | 1,830 | 47 |
| Incoming projected source | 85 | 16,964 | 1,788 | 24 |
| New fused and oriented | 82 | 16,964 | 1,723 | 23 |
| Incoming projected source | 84 | 2,097,732 | 1,775 | 23 |
| New fused and oriented | 81 | 2,097,732 | 1,711 | 22 |

Table 2: The three-variable reduction preserves the matched exact degree.

The incoming growth quartic [2] has 210 variables, 150 squares, 5,282 monomials, and coefficient height $2^{40}$. Our polynomial-center replacement uses two additional variables and two additional squares. After exact integral row reduction, it has height 6,432. The sparse companion preserves a much smaller support at the larger height 277,420. An existing balanced 214-variable quartic has height 91,679, whereas recent 215-variable proposals reach height 2,304. Those alternatives remain useful. The new quartic improves the low-variable, small-coefficient tradeoff; it does not subsume every existing point on the frontier.

The new positive carrier is compared with the exact incoming Carrier65 [4]: 14 positive variables, degree 150, 65 operations. It removes one variable at the cost of four operations and degree 174. Its alternative lowers the degree to 170 with one further operation. Other repository equations have fewer variables, different degrees, or fewer operations. In particular, 81 is not a global variable-count record, and the 39-integer translation of the compact equation is a transparent domain conversion rather than a claim of optimality.

The phrase "simplest equation" needs a chosen metric. Variables, degree, expanded support, coefficient height, circuit size, coefficient-description length, and proof length can move in different directions. This report records those distinctions instead of assigning an unsupported universal ordering. The new ideas are concrete improvements relative to the inspected families. A claim of historical priority across all literature would require a separate investigation.

# 2 The arithmetic and logical dependency boundary

The two direct families use the repository's numerical calculus C222. Their semantic chain is

$$\begin{aligned} &\text{integer zero} \iff \text{finite square-shell C222 refutation} \\ \iff\ &\text{finite original C222 refutation} \iff \text{finite ZF refutation} \\ \iff\ &\text{finite ZFC refutation.} \end{aligned}$$

The square-shell recoding is an explicit recursive bijection on numerical syntax. It fixes truth 0 and falsity 1 and transports the guarded schema. The next two logical endpoints are the identified repository results `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. Their source review and exact provenance are retained in the archive [6]. The direct equations do not require a Turing-machine simulation or Jones' universal polynomial. The separate compact family retains its native coefficient/compiler ancestry, including the inherited machine-to-refutation calibration.

On the arithmetic side, the inherited direct source proves finite Horn-flow extraction, beta-CRT reconstruction, factorial-free binomial selection, and signed Pell index classification. This submission changes three specifically identified arithmetic interfaces and proves their new soundness and completeness. It does not infer those unbounded facts from sample computations.

The proof status has three distinct parts. First, exact finite polynomial identities and source bindings are checked by executable verifiers. Second, the new unbounded arithmetic lemmas are ordinary mathematical proofs, reviewed independently in this work. Third, the C222/ZF/ZFC and native calibrations are credited inherited results. The selected Lean source evidence is not a full newly compiled transitive import closure, and this release makes no new end-to-end Lean or Rocq claim.

All recursions and completeness constructions concern finite internally coded syntax, natural numbers, finite rings, and finite lists. Their induction, integer arithmetic, finite CRT, and Pell arguments are available in ZFC. Consequently the intended biconditional concerns each model's internal integers and finite derivations. It does not impose an external "standard proof code" restriction or assume omega-consistency. The equivalence proves neither Con(ZFC) nor its negation.

# 3 A polynomial Pell center with small coefficients

## 3.1 The growth interface

The square-shell constructors used by the direct source are

$$I(u,v) = 2((u+v)^2+u), \qquad M(u,v) = I(u,v)+1,$$

$$A(u,v) = 2((u+v)^2 + 3u + 4v + 3), \qquad E(u,v) = A(u,v) + 1.$$

The authenticated local axiom terms have depths $4, 3, 3, 3, 3, 5, 4, 3, 19$. Every local leaf parameter is below the horizon $t$. For $0 \le u, v \le m$, the largest constructor satisfies

$$E(m,m) + 1 = 8m^2 + 14m + 8 \le 8(m+1)^2.$$

Induction on depth therefore bounds every local formula label strictly by

$$L = 8^{d-1}t^d, \qquad d = 2^{19}. \tag{2}$$

Indeed the recurrence $H_0 = t$, $H_{j+1} = 8H_j^2$ has solution $H_j = 8^{2^j-1}t^{2^j}$.

Weighted-label separation needs a sufficiently large scale $F$, eventually positive, with $|F| \ge L^t$. The incoming standalone certificate obtains this from a Pell center $4t+1$ and the divisibility condition $F = 2^{19}t(2u+1)$. That large coefficient dominates its collected quartic. We move the growth into the Pell center itself.

## 3.2 Five new coordinates and a cofinal horizon

Retain the front coordinates $t, F$ and impose

$$\boxed{\begin{array}{lll} t = 64 + 4z^2, & S = t^2, & T = S^2, \\ Z = TF, & X^2 - Z^2 - 2ZF = 1, & F = T(2u+1). \end{array}} \tag{3}$$

The five growth witnesses are $S, T, Z, X, u$; $z$ is the existing horizon witness. Each displayed residual is quadratic. Eliminating the polynomial abbreviations in the proof, rather than in the literal quartic, gives

$$X^2 - \big((T+1)^2 - 1\big)F^2 = 1, \qquad T = t^4.$$

Thus the center is $a = t^4 + 1$.

**Lemma 3.1** (Elementary Pell facts)**.** *For $a \ge 2$, define $X_n(a), Y_n(a)$ by*

$$X_0 = 1, \quad Y_0 = 0, \quad X_{n+1} = aX_n + (a^2-1)Y_n, \quad Y_{n+1} = X_n + aY_n.$$

*Every integral solution of $X^2 - (a^2-1)Y^2 = 1$ with $Y \ne 0$ satisfies $(|X|, |Y|) = (X_n(a), Y_n(a))$ for a unique $n \ge 1$. Moreover*

$$Y_n(a) \equiv n \pmod{a-1}, \qquad Y_n(a) \ge (2a-1)^{n-1}.$$

*If $a$ is odd, $\nu_2(Y_n(a)) = \nu_2(n)$.*

*Proof.* For positive $X, Y$ with $Y > 1$, multiply the Pell unit by $a - \sqrt{a^2-1}$. The new ordinate $aY - X$ is positive because $X < aY$, and smaller than $Y$ because $X > (a-1)Y$. The new abscissa $aX - (a^2-1)Y$ is positive by squaring the two positive sides. Descent ends at $(a, 1)$, proving classification and uniqueness. The index congruence follows by evaluating the integer recurrence at $a = 1$. The second-order recurrence and monotonicity give $Y_{n+1} = 2aY_n - Y_{n-1} \ge (2a-1)Y_n$. For odd $a$, every $X_n$ is odd and $Y_n$ has the parity of $n$. The duplication identity $Y_{2n} = 2X_nY_n$ then proves the valuation formula. □

**Theorem 3.2** (Growth without a large fixed multiplier)**.** *Every integer solution of (3) has even nonzero $F$ and*

$$|F| > \big(8^{2^{19}-1}t^{2^{19}}\big)^t = L^t.$$

*In particular $|F| > t$.*

*Proof.* We have $t \ge 64$, $4 \mid t$, and $T = t^4 > 0$. The odd factor in $F = T(2u+1)$ excludes $F = 0$ and makes $F$ even. Apply Lemma 3.1 at $a = T + 1$: $|F| = Y_n(a)$ for $n \ge 1$. Since $T \mid F$ and $Y_n(a) \equiv n \pmod{T}$, the positive index satisfies $T \mid n$, hence $n \ge T = t^4$. Therefore

$$|F| \ge (2t^4 + 1)^{t^4-1} > (2t^4)^{t^4-1}.$$

Put $d = 2^{19}$. Since $t^3 \ge 64^3 = d/2$, $t^4 - 1 > t(d/2 - 1)$. Also

$$(2t^4)^{d/2-1} = 2^{d/2-1}t^{2d-4} > 2^{3d-3}t^d = L.$$

For the last inequality it suffices that $t^{d-4} > 2^{(5/2)d-2}$; already $t \ge 16$ gives the stronger lower bound $t^{d-4} \ge 2^{4d-16}$. Raising to $t$ proves the claim. The argument makes no assumption about the sign of $F$. □

### 3.3 Completeness: the residue condition must survive

A large Pell solution alone would be insufficient. The subsequent selector needs $B = \binom{F+t}{t}$ odd, so the completeness construction must select a large ordinate with a compatible binary residue.

**Lemma 3.3** (Pell permutation modulo powers of two)**.** *If $a \equiv 1 \pmod 4$, then $s \mapsto Y_s(a)$ induces a permutation modulo $2^r$ for every $r \geq 1$. Every residue class is attained at arbitrarily large nonnegative indices; odd residues have odd indices.*

*Proof.* Modulo two, $Y_s \equiv s$. All $X_s$ are $1 \pmod 4$. For $h = 2^r$, the valuation formula gives $\nu_2(Y_h) = r$. The Pell norm and $8 \mid a^2 - 1$ imply $X_h \equiv 1 \pmod{2^{2r+2}}$, because $X_h + 1$ has valuation one. The addition formula then gives

$$Y_{s+2^r} - Y_s = (X_{2^r} - 1)Y_s + X_s Y_{2^r} \equiv 2^r \pmod{2^{r+1}}.$$

The two lifts of each index class give the two distinct residue lifts. Induction proves bijectivity. The same identity gives periodicity at each modulus, and adding periods gives arbitrarily large indices. □

Choose $z$ divisible by $2^{12}$, sufficiently large to accommodate a given finite refutation and its padded arithmetic trace. Such horizons are cofinal, and

$$t \equiv 64 \pmod{2^{26}}, \qquad \nu_2(t) = 6, \qquad \nu_2(T) = 24.$$

In particular the bit of $t$ at position 24 is zero. Set

$$a = T + 1, \quad n_0 = T, \quad y_0 = Y_T(a), \quad a_0 = X_T(a), \quad b_0 = y_0/T.$$

The index congruence makes $b_0$ integral, and the valuation formula makes it odd. Since $T$ is even, the duplication formula gives $a_0 \equiv 1 \pmod 4$. Choose $\ell > 26$ with $t < 2^\ell$. Both $T/2^{24}$ and $b_0$ are invertible modulo $2^{\ell-24}$. Lemma 3.3 supplies an arbitrarily large positive odd $s$ satisfying

$$(T/2^{24})b_0 Y_s(a_0) \equiv 1 \pmod{2^{\ell-24}}.$$

Define

$$F = Y_{Ts}(a) = y_0 Y_s(a_0), \quad X = X_{Ts}(a), \quad Z = TF, \quad u = (F/T - 1)/2.$$

Then $u \in \mathbb{Z}$, every growth row holds, and

$$F \equiv 2^{24} \pmod{2^\ell}. \tag{4}$$

The binary supports of $F$ and $t$ are disjoint: below $\ell$, the only nonzero bit of $F$ is 24, and $t$ has no bits at or above $\ell$. Consequently addition has no carries. The identity

$$\nu_2\binom{F+t}{t} = s_2(F) + s_2(t) - s_2(F+t)$$

shows that $B$ is odd.

The remaining canonical construction now applies. Choose an arbitrarily large odd multiplier containing every odd prime at most $t$ and set $y = F(2v+1)$. Then every prime at most $t$ divides $y$, so the beta moduli $1 + iy$ for $1 \leq i \leq t+1$ are pairwise coprime. This requires prime support, not the stronger assertion $t! \mid y$. Finite CRT supplies the beta code $x$. With $N = B^x$ odd, choose the independent selector representatives first, and choose the even anchor $C$ last, larger than every other actual exponent. Its retained congruence gives $B \mid C + F + 1$. Set the actual radix $D = 2(C + F + 1)$ and $q = D^x$; then $N \mid q$ and $D > 2C$. The inherited digit estimates and main Pell frame complete the remaining witnesses.

### 3.4 The horizon change and the signed bootstrap

Changing $t = 16 + 4z^2$ to $t = 64 + 4z^2$ changes the allowed horizons. It is a semantic cofinal replacement, not a coordinatewise map between every old and new zero. The emitted source audit verifies that the horizon helper occurs only in its defining row. The retained arithmetic uses $t \geq 16$, $4 \mid t$, and the ability to choose a sufficiently large horizon. All are preserved, and the residue construction above establishes the needed new cofinal completeness.

Soundness retains the incoming delayed sign proof. The new growth block first supplies only $|F| > t$ and even nonzero $F$. The even anchor and even $F$ make $D = 2(C + F + 1)$ nonzero. The signed exact-power request $q = D^x$, followed by the beta endpoints, forces $x \geq 1$. The request at exponent $y+1$ then forces $y > 0$; the endpoints further give $x \geq 2 + (t + 1)y$. Now all bases $q + 1$ have magnitude at least three. Their retained requested exponent $F + t$ must be nonnegative, and $|F| > t$ excludes $F < 0$. Thus $F > 0$, and the full separation bound is available. The anchor becomes positive as well. The old radix and binomial decoding therefore proceed in the original noncircular order.

### 3.5 Literal graft and reversible row reduction

The new raw equation is obtained from the pinned principal210 polynomial by changing exactly four old rows and appending two rows. The source coordinates are

$$t = x_0, \quad F = x_2, \quad z = x_{178}, \quad X = x_{207}, \quad Z = x_{208}, \quad u = x_{209}, \quad S = x_{210}, \quad T = x_{211}.$$

The six affected rows are

$$\begin{aligned}
r_{125} &= x_0 - 64 - 4x_{178}^2, \\
r_{147} &= x_{208} - x_2 x_{211}, \\
r_{148} &= x_{207}^2 - x_{208}^2 - 2x_2 x_{208} - 1, \\
r_{149} &= x_2 - x_{211} - 2x_{209} x_{211}, \\
r_{150} &= x_{210} - x_0^2, \\
r_{151} &= x_{211} - x_{210}^2.
\end{aligned}$$

The other 146 rows are retained literally. This gives $P_{212}^{\text{sparse}} = \sum r_j^2$.

The small-coefficient version is obtained by 465 elementary integral row operations $r_i \leftarrow r_i + k r_j$, $i \neq j$. Reverse the order and negate $k$ to invert them. Therefore the transformed rows vanish simultaneously exactly when the original rows do. Their sums of squares have exactly the same integer zero set. No variable is added or removed in this step, and no sign or divisibility condition is introduced.

The independent checker collects the complete quartics and obtains

| | monomials | height | largest residual coefficient |
|---|---|---|---|
| $P_{212}^{\text{sparse}}$ | 5,290 | 277,420 | 226 |
| $P_{212}^{\text{small}}$ | 21,556 | 6,432 | 60. |

For example the collected coefficient of $x_8 x_{10} x_{23} x_{31}$ in $P_{212}^{\text{small}}$ is 6,432. It both witnesses the height and proves that the degree is exactly four. The denser support is an explicit cost of the smaller coefficients.

## 4 One certificate across two arithmetic stages

The direct source has two distinct uses for nonnegativity. A Pell cap bounds all classified indices before their signs are recovered. A common digit budget selects the canonical representatives of ten binomial remainders. Both were represented by the universal ternary polynomial

$$\mathcal{G}(a, b, c) = a^2 + b^2 + c(c + 1).$$

For integer inputs this is nonnegative, and it represents every nonnegative integer. Indeed, the three-square theorem applied to $4n + 1$ gives one odd and two even roots; division by four gives the displayed representation.

The new construction shares the certificate across these two stages. It uses the stronger budget to recover both original inequalities before either arithmetic stage is interpreted.

### 4.1 A general lemma

Let $U, V \geq 0$, $E \geq 1$, and $R \geq 0$ be integers, and put

$$H = UV - UE - VR - 1.$$

If $H \geq 0$, then $U, V > 0$: setting either to zero makes $H < 0$. The identities

$$U(V - E) = VR + 1 + H > 0, \qquad V(U - R) = UE + 1 + H > 0$$

give $V > E$ and $U > R$. Thus

$$V - E \geq 0, \qquad U - R - 1 \geq 0. \tag{RG1}$$

Conversely, with positive $U, V$,

$$\frac{H}{UV} = 1 - \frac{E}{V} - \frac{R}{U} - \frac{1}{UV}. \tag{RG2}$$

Consequently one equation $H = \mathcal{G}(a, b, c)$ can replace two separate ternary certificates for (RG1), if the completeness construction satisfies the joint bound in (RG2). The $-1$ excludes the simultaneous zero-scale case, so it must be retained.

## 4.2 The literal application

Here the source subscripts are those of the frozen 219-coordinate arithmetic source retained in the archive. Define

$$C = 2x_{16}, \quad Q = x_{126}, \quad Y = x_{184}, \quad U = Q^2, \quad V = Y^2,$$

and let

$$\begin{aligned}
(y_i)_{i=1}^{13} &= (x_{142}, x_{151}, x_{154} - 1, x_{157}, x_{160}, x_{163}, x_{166}, x_{169}, x_{172}, x_{175}, x_{178}, x_{181}, x_{187}),\\
(r_j)_{j=1}^{10} &= (x_{104}, x_{108}, x_{111}, x_{114}, x_{117}, x_{121}, x_{124}, x_{128}, x_{132}, x_{135}),\\
E &= C^2 + 1 + \sum_{i=1}^{13} y_i^2, \qquad R = 4\sum_{j=1}^{10} r_j^2.
\end{aligned}$$

The translated ordinate $x_{154} - 1$ is essential. Exact coefficient comparison verifies that source rows 150 and 151 are

$$V - E - \mathcal{G}(x_{207}, x_{208}, x_{209}), \qquad U - R - 1 - \mathcal{G}(x_{210}, x_{211}, x_{212}).$$

All six guard coordinates occur only in these two rows. Delete $x_{210}, x_{211}, x_{212}$, reuse the other triple, and replace both rows by

$$\boxed{UV - UE - VR - 1 - \mathcal{G}(x_{207}, x_{208}, x_{209}) = 0.} \tag{RG3}$$

Every other residual is retained. The finite source-coordinate reconstructions supplied by the parent are substituted into these expressions; their gates are abbreviations, not additional variables.

## 4.3 Soundness

The nonnegativity premises of the general lemma follow immediately from the displayed sums of squares, without Pell classification or digit decoding. A zero of (RG3) therefore supplies both old inequalities. Represent their nonnegative slacks by two fresh ternary triples and restore the old six guard coordinates. All other rows remain valid, because those coordinates are private. The certified acyclic unit-pivot reconstruction restores the parent's eliminated coordinates. Its Pell cap and subsequent sign bootstrap can then be invoked in their original order. This proves that every new zero yields a finite syntactic refutation under the identified inherited calibration.

## 4.4 Completeness with a quantitative margin

Starting from a finite refutation, the inherited canonical construction chooses a positive even anchor $C$ strictly larger than every other principal index $k_i$, and then a positive main Pell center $A$ with

$$W \geq 258, \qquad A > W^W, \qquad C^2 < W.$$

It supplies $Y = Y_C(A)$, $y_i = Y_{k_i}(A)$, $Q = q^x \geq 2$, and the ten normalized remainders $0 \leq r_j < Q/32$. Write $Z = Y_{C-1}(A) \geq 1$. The recurrence and monotonicity give

$$Y_C(A) \geq (2A - 1)Z, \qquad y_i \leq Z.$$

Therefore

$$\frac{E}{V} \le \frac{W+13Z^2}{(2A-1)^2Z^2} \le \frac{W+13}{(2A-1)^2} < \frac{1}{2}, \qquad \frac{R}{U} < \frac{40}{32^2} = \frac{5}{128}.$$

The first strict inequality follows already from $A > W \ge 258$. Also $C \ge 2$, so $Y_C(A) \ge Y_2(A) = 2A \ge 4$. Hence $U \ge 4$, $V \ge 16$, and

$$\frac{E}{V} + \frac{R}{U} + \frac{1}{UV} < \frac{1}{2} + \frac{5}{128} + \frac{1}{64} = \frac{71}{128} < 1.$$

Thus $H > 0$. Its ternary representation supplies the new triple. The CRT representatives and powers are chosen first, the main Pell frame after them, and this final representation last. There is no circular witness choice.

This is an equivalence of existence of zeros, established by canonical-witness completeness. It is not a pointwise bijection between every old zero and every new zero.

## 4.5 Composing the linear orientation numerator

The independent orientation improvement replaces

$$\Omega_0 = \sum_i \big((X_i - 1)^2 + (Y_i - k_i)^2\big) \quad \text{by} \quad \Omega_1 = \sum_i \big(1 - X_i + (Y_i - k_i)^2\big).$$

Under the bounded raw Pell classification, $(X_i, Y_i) = \epsilon_i(X_{\nu_i}(A), Y_{\nu_i}(A))$, reduction modulo $A - 1$ gives the nonnegative error

$$\sum_i \big(1 - \epsilon_i + (\epsilon_i\nu_i - k_i)^2\big).$$

It vanishes precisely when every $\epsilon_i = 1$ and $\nu_i = k_i$. The same signed rational-integrality argument authenticates this error simultaneously with all power requests. The complete bound and restoration proof are retained in the accompanying orientation proof. The two changes concern disjoint private rows and therefore compose.

The recommended resulting polynomial has 82 unrestricted integer variables, 23 residual squares, exact degree 16 964, and 1 723 live binary arithmetic operations. Its lower-dimensional companion has 81 variables, 22 residual squares, degree 2 097 732, and 1 711 operations. A lower-degree companion has 106 variables, degree 32, 47 residual squares, and 1 830 operations. All three retain maximum absolute literal 1 099 511 627 780; this is not the expanded coefficient height. Conservative height bounds remain $2^{59\,805}$, $2^{7\,342\,557}$, and $2^{308}$, respectively.

Exact sparse identities bind the source predicates, structural substitution checks bind all retained expressions and unit pivots, and nonzero top homogeneous coefficients establish the stated degrees. Nonzero derivative certificates establish genuine dependence on every input. These finite certificates complement the unbounded proof; the consistency equivalence still uses the explicitly identified inherited C222/ZF/ZFC calibration.

## 4.6 Full bounded-error argument for the linear orientation

Here is the unbounded estimate underlying the orientation substitution. It applies at the raw-classification stage, before requested indices have been authenticated. Put $D = A - 1$, assume $|A| > W^W$, $W \ge 258$, and write $b = \lfloor\sqrt{W-1}\rfloor$. The retained anchor and comparison rows classify at most 15 principal pairs as

$$(X_i, Y_i) = \epsilon_i(X_{\nu_i}(A), Y_{\nu_i}(A)), \quad \epsilon_i \in \{1, -1\}, \quad |\nu_i|, |k_i| \le b.$$

Signed-index Pell polynomials satisfy $X_n(A) \equiv 1$ and $Y_n(A) \equiv n \pmod D$. Therefore

$$\Omega_1 \equiv o := \sum_i \big(1 - \epsilon_i + (\epsilon_i\nu_i - k_i)^2\big) \pmod D.$$

Every summand of $o$ is nonnegative, and it vanishes exactly at $\epsilon_i = 1$, $\nu_i = k_i$. Moreover $0 \le o \le 90b^2$. In the standalone orientation certificate this bound already proves that $D \mid \Omega_1$ is equivalent to the desired canonical orientation. Notice that $\Omega_1$ itself need not be nonnegative; its bounded residue $o$ has the required sign.

For joint power authentication, let $n_j$ be the at most five bases, and let $(m_i, k_i)$ be their at most seventeen requests. Every base, requested output, and index has absolute value at most $b$. Define

$$T_j = 2An_j - n_j^2 - 1,$$
$$E_j = \sum_{i \in j} \big((m_i - X_i + AY_i)^2 - Y_i^2 - 2n_j Y_i (m_i - X_i)\big).$$

The literal joint residual is

$$\Omega_1 \prod_j T_j + D \sum_j n_j E_j \prod_{\ell \neq j} T_\ell - zD \prod_j T_j = 0. \tag{5}$$

All denominators in its mathematical interpretation are nonzero. A zero base gives $T_j = -1$; otherwise $|T_j| > W^W$ and $n_j/T_j$ has the sign of $A$. Also $D$ has the sign of $A$ and $|D| \geq W^W$.

For clarity, the relevant power congruence follows from the identity

$$(m - X + AY)^2 - Y^2 - 2nY(m - X) = (m - X + (A - n)Y)^2 + (2An - n^2 - 1)Y^2.$$

Modulo $T = 2An - n^2 - 1$, the expression $X_\nu(A) + (n - A)Y_\nu(A)$ is $n^\nu$; this follows from its recurrence, starting at $1, n$. For negative indices this is a modular inverse, permitted because $\gcd(n, T) = 1$. Equivalently one may clear all powers before stating the congruence.

Use the latter form by taking the mathematical clearing integer

$$B_* = \prod_{j : n_j \neq 0} n_j^b, \qquad \delta_i = B_* m_i - \epsilon_i (B_* / n_j^b) n_j^{b + \nu_i}.$$

These are integers since $b + \nu_i \geq 0$. The congruences and (5) imply that

$$\frac{B_*^2 o}{D} + \sum_{j : n_j \neq 0} \sum_{i \in j} \frac{n_j \delta_i^2}{T_j}$$

is an integer. Every nonzero summand has the same sign, that of $A$. Since $|B_*| \leq b^{5b}$ and $|\delta_i| \leq 2b^{6b}$, its total absolute numerator estimate is

$$B_*^2 o + \sum_i |n_j| \delta_i^2 \leq 90 b^{10b+2} + 68 b^{12b+1} \leq 158 b^{12b+2} < W^{6b+2} \leq W^W.$$

Here $b \geq 16$, so $158 < b^2$, and $6b + 2 \leq W$. Dividing by the lower denominator bound shows that the displayed integer has absolute value strictly below one. It must be zero. Its same-sign terms vanish separately, giving $o = 0$ and every $\delta_i = 0$. Thus the indices and signs are canonical and every nonzero-base request is the exact signed power. In particular a base of absolute value at least two cannot have a negative requested exponent with an integral output.

The source's zero-base bootstrap is retained until the front equations exclude those temporary cases. Conversely, canonical pairs make $D \mid \Omega_1$, and exact powers make $T_j \mid E_j$ for nonzero bases; zero-base denominators are units. Their quotient sum supplies the new $z$. This proves completeness without a pairwise-coprimality assumption on the different $T_j$. The entire argument uses bounded classified indices before canonicality, so it does not assume its own conclusion.

## 5 Two Pell roots carry two quotients

This section gives an independent compact improvement, retaining the exact admissible942 coefficient tuple of the incoming *Pell Carrier65* package. That package removes the first-index quotient by using the first Pell root. We now use a second Pell root to remove the native transport quotient. The factor four in an odd-ordinate congruence is what makes the second reduction work without a parity restriction on the auxiliary base. The resulting ordinary polynomial has thirteen strictly positive variables, 69 live binary arithmetic operations, and exact degree 174. Its standard unrestricted-integer presentation has 39 variables, 160 operations, and degree 348. A companion trades one extra operation for degrees170 and340.

The parent and its current calibration are retained under the compact component's `inherited/` directory. In particular the complete admissible942 proof, rather than an arbitrary four-number tuple satisfying elementary inequalities, gives these equations their consistency meaning. The general use of a Pell ordinate as a quotient carrier already occurs in package 170. The present contribution is the stronger residue, its use for native sign recovery before rank decoding, and its composition with the first-root construction.

## 5.1 The entire polynomial

There are exactly thirteen independent positive integer inputs, in the order used by the machine-readable equation:

$$J, F, \alpha, f, i, T, s, w, n, \eta, \zeta, m, \gamma > 0.$$

Retain the four fixed integers $\beta, K_0, M_C, M_F$ from the exact parent. Their unsigned big-endian byte strings total 94,689 bytes, and their largest bit length is 189,693. The compact component includes their complete finite reconstruction, its 399-character seed, and all four literal byte strings. The polynomial below contains no variable exponent, division, floor, factorial, or unspecified predicate. Every other letter is a polynomial abbreviation:

$$\begin{aligned}
h &= \beta J, & q &= h + 1, & C &= q - F - \alpha,\\
X &= wq, & Y &= sq^3, & E &= XY,\\
k &= \eta + \zeta, & \ell &= EkY, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1,\\
D &= X + ac + \gamma H, & \tau &= En + k - R, & B &= (K_0 + w)C - F,
\end{aligned}$$

where

$$R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J.$$

The last definitions are

$$\begin{aligned}
b &= cT - R, & V &= f^2 b - c, & S &= hi\Delta c^2,\\
L &= V^2, & y &= 1 + 4\big[h(m - LB) - B\big],\\
N_0 &= \tau^2 - \ell(\ell + k), & N_1 &= D^2 - \Delta c^2,\\
N_a &= S^2(V^2 - y^2) + y^2, & N_s &= \Delta f^2 - S^2.
\end{aligned}$$

The single equation is

$$\boxed{P = N_0 N_1 N_a N_s - \Delta = 0.} \tag{6}$$

The companion changes only $L$ to $bV$. Notice that $y$ is a signed polynomial value, not an additional positive unknown.

## 5.2 A residue with a factor of four

For an integer $S \geq 2$, write

$$\chi_S(r) + \psi_S(r)\sqrt{S^2 - 1} = (S + \sqrt{S^2 - 1})^r.$$

The odd subsequence $u_j = \psi_S(2j + 1)$ satisfies

$$u_0 = 1, \quad u_1 = 4S^2 - 1, \quad u_{j+2} = (4S^2 - 2)u_{j+1} - u_j.$$

Induction modulo $4S^2$ therefore proves

$$\psi_S(2j + 1) \equiv (-1)^j \pmod{4S^2}. \tag{7}$$

This holds for odd as well as even $S$. The weaker congruence modulo $S^2$ would be insufficient for the sign selection used below.

Suppose that integers $V, y$ satisfy

$$S^2V^2 - (S^2 - 1)y^2 = 1. \tag{8}$$

Neither coordinate is zero. The positive solutions of $x^2-(S^2-1)u^2=1$ are exactly $(\chi_S(r),\psi_S(r))$, $r\geq 1$. One can prove this directly by descent: for $u>1$, multiplication by $S-\sqrt{S^2-1}$ sends the ordinate to $Su-x$, which is positive and smaller than $u$, because $S-1<\sqrt{S^2-1}<x/u<S$. The new abscissa is positive as well. The terminal positive pair is $(S,1)$, and reversing the descent gives the stated powers.

Apply this to $x=S|V|$ and $u=|y|$. Modulo $S$, the even-index $\chi$ values are $+1$ or $-1$, whereas the odd-index values are zero. The index is consequently odd. Equation(7) then gives

$$y\equiv +1 \text{ or } -1 \pmod{4S^2}.$$

If $y\equiv 1 \pmod 4$, the minus sign is impossible. Thus

$$y\equiv 1 \pmod 4 \quad\Longrightarrow\quad 4S^2 \mid y-1. \tag{9}$$

Also, directly from(8),

$$|y|<2|V|. \tag{10}$$

No packed-rank interpretation or positivity of $V$ or $y$ was used.

### 5.3 Soundness before rank recovery

On the entire positive domain, $h\geq 31$, $c>1$, $\Delta>1$ and $S>c$. At a zero of(6), cancellation of the positive integer $\Delta$ gives

$$N_0N_1N_a\left[f^2-\Delta(hic^2)^2\right]=1.$$

All four factors are integer units. Each negative sign is excluded independently. The consecutive-product negative-Pell obstruction excludes $N_0=-1$, since $\ell(\ell+k)=EY(EY+1)k^2$ and $EY\geq 2$. For completeness, if $x^2-t(t+1)u^2=-1$ with $t\geq 2$ and least positive $u$, then $tu<x<(t+\frac{1}{2})u$ after taking absolute values; multiplication by $(2t+1)-2\sqrt{t(t+1)}$ gives a positive smaller ordinate $(2t+1)u-2x$, a contradiction. This does not assume $\tau>0$.

Because $\Delta$ is zero or three modulo four, neither $N_1$ nor the normalized strong norm can be $-1$. Finally $N_a$ is congruent modulo four to either $y^2$ or $V^2$, and cannot be $-1$. Hence

$$N_0=N_1=N_a=1, \qquad N_s=\Delta. \tag{11}$$

In particular $N_a=1$ is(8). Our definition of $y$ gives $y\equiv 1 \pmod 4$, so (9), together with $h\mid S$, yields

$$h^2 \mid h(m-LB)-B, \qquad \text{and therefore } h\mid B.$$

No primality or parity assumption on $h$ enters this deduction.

For the main profile, $L=V^2\geq|V|\geq 1$. For the companion $L=bV$, the same inequality holds at a zero. Indeed $b=0$ would give $V=-c$ and

$$y^2-c^2=\frac{c^2-1}{S^2-1}\in(0,1),$$

impossible for integers. Equation(11) gives $f^2=1+\Delta h^2i^2c^4>c$, so $V=f^2b-c$ has the sign of nonzero $b$. Therefore $bV=|b||V|\geq|V|$.

Set $W=1+hL>0$. If $B\leq -1$, positivity of $m$ forces

$$y=1+4(hm-WB)\geq 1+4(h+W)>4hL\geq 4|V|,$$

contradicting(10). Thus $B\geq 0$. The omitted native quotient is consequently recovered as the positive integer

$$z=B/h+1.$$

Its parent factor is exactly

$$N_t=(K_0+w)C+(q-F)-zh=B+q-zh=1.$$

Keep the other parent inputs, replace its strong multiplier by $hi$, and replace its positive auxiliary ordinate by $|y|$. Every parent factor now has the required value, and the Carrier65 polynomial vanishes. This is the complete soundness map. It uses no conclusion about $C$, $R$, or the first Pell index before reaching the already proved parent.

## 5.4 Completeness with a fresh strong seed

A finite syntactic ZFC refutation supplies the parent's canonical outer data. Among their properties are

$$N_0 = N_1 = 1, \quad R \equiv 3 \pmod 4, \quad c = \psi_{a+2}(R), \quad X = 2^R,$$

as well as $C > 0$, $F < q$, $h \mid B$, and $R > 3q$. The inherited first-root coordinate $n$ is positive. These data are independent of the private strong seed and the auxiliary coordinates.

There is actually a strict native surplus: $2^R > q^2$ gives $w = X/q > q$, and hence

$$B = (K_0 + w)C - F > w - q > 0. \tag{12}$$

Choose a positive solution

$$f^2 - \Delta I^2 = 1, \qquad hc^2 \mid I.$$

Such solutions exist cofinally. In the finite ring $(\mathbb{Z}/(hc^2)\mathbb{Z})[t]/(t^2 - \Delta)$, the element $a + 2 + t$ is a unit of norm one and therefore has finite order. Any positive multiple of that order gives $I \equiv 0 \pmod{hc^2}$; its positive powers give arbitrarily large choices. Set $i = I/(hc^2) > 0$ and $S = \Delta I$. Then $N_s = \Delta$ for our definitions.

Put

$$V_* = \frac{\chi_S(R)}{S}, \qquad y_* = \psi_S(R).$$

The quotient is a positive integer because $R$ is odd. Write $R = 2j + 1$ and $\chi_S(R)/S = Q_j(S^2)$. The elementary identities retained in the parent proof are

$$Q_j(0) = (-1)^j R, \qquad Q_j(1 - A^2) = (-1)^j \psi_A(R), \quad A = a + 2.$$

Since $j$ is odd, $S^2 = 1 - A^2 \pmod{f^2}$ and $c \mid S$, they give $V_* \equiv -c \pmod{f^2}$ and $V_* \equiv -R \pmod c$. Moreover $\gcd(c, f) = 1$ and $f^2 \equiv 1 \pmod c$. Thus

$$T = \frac{V_* + c + Rf^2}{cf^2} > 0$$

is an integer and our polynomial value $V$ is $V_*$. These quotients construct witnesses; none is an operation in the delivered polynomial.

Equation(7) gives $y_* \equiv -1 \pmod{4S^2}$. Use the signed value $y = -y_*$, so $y \equiv 1 \pmod{4S^2}$, and define

$$m = \frac{y - 1}{4h} + \frac{(1 + hL)B}{h}. \tag{13}$$

It is an integer. It is positive because $B$ is a positive multiple of $h$ by(12), $L \geq V \geq 1$, and $|y| < 2V$:

$$m > -\frac{2V + 1}{4h} + 1 + hL > 0.$$

Equation(13) makes the literal carrier equal the chosen signed $y$, and all four factors in (11) have their required values. Thus $P = 0$. This is an existence construction that refreshes the private Pell seed; it is not a bijection on every old and new zero.

## 5.5 Exact size and unrestricted integer variables

Relative to the 65-operation parent, remove the three final native-factor rows, add the one subtraction defining $B$, remove one product multiplication, scale the strong multiplier by $h$ using one multiplication, and compute $y$ with six operations. The count is therefore $65 - 3 + 1 - 1 + 1 + 6 = 69$. The companion requires one extra multiplication for $bV$. Squares and scalar multiplications are charged as binary operations.

The respective factor degrees are

| | $N_0$ | $N_1$ | $N_a$ | $N_s$ | $P$ |
|---|---|---|---|---|---|
| $L = V^2$ | 22 | 18 | 86 | 48 | 174 |
| $L = bV$ | 22 | 18 | 82 | 48 | 170 |

The main-norm degree uses the exact cancellation, with $Z = X + \gamma H$,

$$(ac + Z)^2 - (a^2 + H)c^2 = 2acZ + Z^2 - Hc^2.$$

Let $q_0 = \beta J$ and $k_0 = \eta + \zeta$. The complete leading homogeneous form of the main equation is

$$-128 q_0^{99} \gamma i^4 k_0^{15} w^{14} s^{27} f^8 T^4 (q_0 - F - \alpha)^2.$$

The companion replaces $f^8$ by $f^4$. The coefficient selecting $\eta^{15}$ and $F^2$ is $-128\beta^{99}$, which is nonzero on the actual fixed tuple. These are exact degrees, not only graph estimates.

To obtain unrestricted integer equations, replace each positive input by

$$1 + u^2 + v^2 + w(w+1), \qquad u, v, w \in \mathbb{Z}.$$

The classical three-square theorem applied to $4N + 1$ proves that every $N \geq 0$ is $u^2 + v^2 + w(w+1)$: exactly one of the three squares is odd, and subtracting its1 and dividing by4 gives the representation. Conversely this expression is nonnegative on all integer triples. The substitution uses seven operations per coordinate. It gives 39 integer variables and respectively 160 or 161 operations. Degrees double to348 and340. Nonzero highest homogeneous terms survive: setting each second and third new coordinate to zero doubles the monomial exponents injectively.

The independent compact checker verifies 25 or 26 exact formula cuts, the full 96-term leading forms, all coordinate and coefficient bindings, the actual fixed-tuple degree witnesses, four coefficient reconstructions, and 1,278 finite Pell-carrier fixtures. Targeted changed gates are rejected. These finite checks complement the preceding unbounded proof. They do not replace the inherited native/K215/refutation calibration or provide a fresh end-to-end proof-assistant derivation. Composing that calibration with the two witness maps and the exact domain translation proves in ZFC that the integer equation has no zero if and only if Con(ZFC).

The result is a tradeoff against the matched parent and inspected compact family. Package 170 has a 13-positive-variable, 70-operation degree 206 point and a 71-operation degree 162 point, with a different 295,388-byte tuple. The present 69-operation degree 174 point does not assert a minimum over all Diophantine equations, all integer domains, or all coefficient architectures.

# 6 Reproduction, evidence, and integration

## 6.1 Files that define the equations

The archive is organized around three independently usable components:

- `quartic212/` contains both literal 212-variable systems, their collected quartics, the frozen principal210 input, the source-change certificate, the reversible row ledger, builders, evaluator, and an independent checker.
- `projected/` contains the 106-, 82-, and 81-variable direct graphs, their exact input and reconstruction maps, frozen parent graphs, the immediate parent archive, and separate construction and verification programs.
- `compact13/` contains the two 13-positive-variable graphs, their 39-integer lifts, the fixed coefficient tuple and its reconstruction, the exact incoming Carrier65, and ordinary proofs and checkers.

The top-level `verify.py` runs the supplied finite checks. It does not search for roots. The checksum manifest authenticates the delivered bytes. The report source and all included equation definitions are editable.

## 6.2 What was actually checked

For the quartic, a checker independent of the producer verifies the pinned source hash, all 146 unchanged residuals, the six displayed affected rows, all 212 live coordinates, and the horizon helper's sole occurrence. It replays all 465 row operations forward and backward and independently collects both polynomials. Its seven targeted corruption checks reject changed growth coefficients and a changed row operation. Pell recurrence fixtures are auxiliary tests only.

For the projected equations, source predicate identities identify the two old guards, structural substitutions identify every retained residual, and acyclic unit-pivot certificates reconstruct the eliminated coordinates. The

linear-orientation replacement is checked as an exact residual change, with a separate ordinary proof of existential equivalence. Nonzero homogeneous terms certify the exact degrees; derivative certificates establish dependence on each listed input. Rebuilding the component reproduces its new JSON artifacts byte for byte.

For the compact equations, an independent checker verifies the formal cuts in the evaluator, the full 96-term leading homogeneous forms, the actual fixed coefficient bytes, and source bindings. The report gives explicit nonzero leading coefficients after specialization. The 39-integer lifts are checked substitutions of the positive parametrization. All substantive verifiers retain their checks under Python's `-O` mode; none relies on disabled assertions for correctness.

No finite check asserts that one of these consistency equations has a root. An evaluation of a test tuple, or a modular recurrence fixture, is an arithmetic test and does not constitute a ZFC refutation. The unbounded theorems remain the mathematical arguments given here and in the named inherited sources.

### 6.3 Suggested repository placement

This package can be incorporated under the repository's `Computability/DiophantineConsistency/` subtree, in its `con-zfc-diophantine-forms/` collection. The directory name should be assigned by the repository's intake process, since numbering may have advanced after the pinned snapshot. The supplied `INTEGRATION.md` records the recommended claims and their qualification. The work creates a reviewable submission and does not modify the remote repository.

## 7 Further research questions

1. **Combine the new growth center with the projected family.** The quartic construction and the 82-variable construction currently descend from different exact immediate sources. A fully authenticated composition could replace the projected family's large growth literals, then rerun its unit-pivot search. The required work is a source-port and frame audit, followed by degree certificates. Counts cannot be obtained safely by merely subtracting the separate savings.
2. **Reduce the coefficient cost of the quartic without excessive support growth.** The 465-step row reduction has a clear certificate but optimizes a quadratic coefficient-vector norm rather than final support or expanded coefficient height. Search over small integral row operations, with a lexicographic or Pareto objective, may improve the 212-variable tradeoff. A further reduction must publish its reversible ledger and recollected polynomial.
3. **Generalize the shared-bound certificate.** The expression $UV - UE - VR - 1$ combines two relative deficits because the canonical completeness construction has substantial spare margin. For several positive scales $U_i$ and nonnegative deficits $E_i$, clearing denominators in $1 - \sum_i E_i/U_i$ suggests further pooling. The main challenge is to exclude all zero-scale cases and recover each original inequality before any dependent decoding. The resulting degree and gate cost may outweigh the saved guard variables.
4. **Lower degree after the quotient deletion.** The 13-positive-variable carrier removes one quotient using a second Pell ordinate, but its positive shift raises the degree. The $bV$ companion already reduces degree by four. A different positive multiplier bounded below by $|V|$, with a lower total degree and no new quantified coordinate, could improve this point. Soundness must handle the sign of $b$ and the case $b = 0$ before using the native decoder.
5. **Formalize the new arithmetic lemmas and seal the full bridge.** The general fused-bound lemma, the odd-Pell congruence modulo $4S^2$, and the polynomial-center growth lemma are compact formalization targets. Afterward, bind the emitted coefficient files to a closed theorem about the fixed object-language ZFC proof predicate. The existing ordinary calibration and finite checker receipts are useful inputs to that work; they do not themselves constitute the completed new formal endpoint.

## References

[1] V. Reshetnikov and contributors, *ProveIt*, pinned revision `b42c796d1f5de9912a576bc2fd77b778ae7402bf`. Repository arithmetic directory.

[2] *Pell growth from divisibility: an explicit quartic for Con(ZFC)*, incoming `ConZFC_Pell_Growth_210_2026-10-25.zip`, same snapshot. The exact principal210 JSON has SHA-256 `20a2108353a3756ed7d1ddfdc71695131dcf0b6d94039988d9565ac48e6e1f0a`. Selected ordinary proofs and attribution are reproduced in `quartic212/upstream/`.

[3] *Direct integer elimination*, incoming `ConZFC_Direct_Integer_Elimination_2026-10-25.zip`, same snapshot. The exact immediate archive is retained in `projected/sources/parent_archive.zip`.

[4] *A first Pell carrier: 65 operations*, incoming `ConZFC_Pell_Carrier65_2026-10-25.zip`, same snapshot. Its exact admissible942 coefficient tuple and carrier65 input are retained with the compact component.

[5] ProveIt packages 135 (cofinal anchors), 151 (factorial-free binomial CRT), 157 (linear CRT), 173 (CRT index anchor), and 176 (oriented pooling). These supply the inherited arithmetic interfaces cited in the respective component proof maps. Exact source hashes and retained paths appear in the provenance files.

[6] ProveIt, `KEndpoint.lean` and `ConstructibleChoice.lean`, identified calibration endpoints. See the retained ordinary dependency audit for their pinned URLs, proof terms, immediate dependencies, and source-review limitations.

[7] P. Pollack and P. Schorn, *Dirichlet's proof of the three-square theorem: an algorithmic perspective*, Mathematics of Computation 88 (2019), no. 316, 1007–1019, doi:10.1090/mcom/3349. The only representation fact used here is the standard consequence $n = a^2 + b^2 + c(c+1)$ for $n \geq 0$; its reduction to three squares is given in the text.

# A The complete small-coefficient quartic

All 212 coordinates $x_0, \dots, x_{211}$ range independently over $\mathbb{Z}$. The equation is $\sum_{j=0}^{151} R_j^2 = 0$. The following literal rows are generated from `quartic212/reduced/equation.json`; a separate parser verifies every printed coefficient and variable index against that file. Their semantic names before row reduction are in the variable table supplied with the equation. Row reduction changes the residual basis but does not change the coordinate domain.

$$R_0 = x_0 - x_0x_4 - x_4 + x_7$$

$$R_1 = -x_2 - x_4x_5 + x_6x_7$$

$$R_2 = -1 + x_1 - x_6x_7$$

$$R_3 = 1 + x_2 - x_3x_{18} + 2x_{16}$$

$$\begin{aligned} R_4 = {} & -1 - x_1 + x_6x_7 - 3x_8 - x_8^2 \\ & - x_8x_9 - x_8x_{10} - 4x_9 + x_9x_{10} + 2x_{10}^2 \\ & + x_{19} - x_{23} - x_{71} \end{aligned}$$

$$\begin{aligned} R_5 = {} & 2x_8 - 2x_8^2 - 8x_8x_{20} - 8x_{20} - 8x_{20}^2 \\ & + x_{21} + x_{33} \end{aligned}$$

$$R_6 = 1 + 2x_{21} - 4x_{21}^2 + x_{22}$$

$$\begin{aligned} R_7 = {} & -1 - x_1 + x_6x_7 - x_9^2 - 2x_9x_{10} \\ & - x_{10}^2 + x_{23} \end{aligned}$$

$$R_8 = 1 - x_8^2 - 2x_8x_{10} - x_{10}^2 + x_{24}$$

$$\begin{aligned} R_9 = {} & 2 + x_1 - x_6x_7 - 13x_8 + 8x_8^2 \\ & + 4x_8x_9 - 14x_8x_{10} - 16x_8x_{19} - 4x_8x_{20} - 12x_8x_{23} \\ & - 24x_8x_{24} + 4x_8x_{52} + 4x_9 - 3x_9^2 + 2x_9x_{10} \\ & + 8x_9x_{23} - 32x_9x_{36} - 38x_{10} - 5x_{10}^2 + 2x_{10}x_{11} \\ & - 52x_{10}x_{19} + 8x_{10}x_{24} + 4x_{10}x_{27} - 24x_{10}x_{36} + 8x_{10}x_{52} \\ & + x_{11}^2 - 17x_{19} + 16x_{19}x_{20} - 8x_{19}x_{23} - 24x_{19}x_{24} \\ & - 8x_{19}x_{27} - 6x_{20} - 4x_{20}^2 - x_{23} + 4x_{23}^2 \\ & + 16x_{23}x_{24} - 8x_{23}x_{36} - 29x_{24} + 4x_{24}^2 + x_{25} \\ & - 12x_{25}^2 + 3x_{26} + 10x_{27} + x_{28} + x_{31} \\ & + x_{33} + 4x_{36} - 4x_{36}^2 + x_{37} - x_{39} \\ & - x_{42} - 8x_{43} + 4x_{52}^2 - x_{53} - x_{59} \\ & + 12x_{71} \end{aligned}$$

$$\begin{aligned} R_{10} = {} & -1 - x_1 + x_6x_7 + 10x_8 - 2x_8^2 \\ & - 6x_8x_9 - 2x_8x_{10} - 8x_8x_{23} + x_9^2 + 3x_{10}^2 \\ & - x_{23} - 4x_{25}^2 + x_{26} - 2x_{43} - 2x_{71} \end{aligned}$$

$$\begin{aligned} R_{11} = {} & 2x_8^2 + 4x_8x_9 + 3x_9^2 + 2x_9x_{10} - 6x_{10}^2 \\ & + 2x_{10}x_{11} - 16x_{10}x_{23} - 16x_{10}x_{24} + x_{11}^2 - 3x_{23} \\ & - 4x_{23}^2 - 8x_{23}x_{24} - 2x_{24} - 4x_{24}^2 + x_{27} \\ & - x_{39} - 4x_{71} \end{aligned}$$

$$
\begin{aligned}
R_{12} = \; & -2 - 2x_1 + x_2 + x_4x_5 + x_6x_7 \\
& + 3x_8 + 3x_8^2 + 6x_8x_{10} + 4x_9 + 23x_{10} \\
& + 7x_{10}^2 - 2x_{10}x_{11} - 16x_{10}x_{19} + 8x_{10}x_{24} + 4x_{10}x_{27} \\
& - x_{11}^2 + 9x_{19} - 16x_{19}x_{24} - 8x_{19}x_{27} - x_{20} \\
& + x_{23} + 22x_{24} + 12x_{27} + x_{28} + x_{39} \\
& - x_{42} + 2x_{71} - x_{131}^2 - x_{131}x_{136} - x_{131}x_{161} \\
& - x_{133}^2 + x_{134}^2 - x_{135} + x_{137} - x_{160} \\
& + x_{162}
\end{aligned}
$$

$$
\begin{aligned}
R_{13} = \; & 4 + x_0 + 3x_1 - 2x_2 - 2x_2x_{208} \\
& - x_3^2 + x_3x_{18} + 2x_3x_{131} - x_4x_5 - 2x_6x_7 \\
& + 10x_8 + 23x_8^2 + x_8x_9 - 3x_8x_{10} + 24x_8x_{19} \\
& + 8x_8x_{23} - 15x_9 - 25x_9^2 + 9x_9x_{10} - 8x_9x_{23} \\
& - 4x_9x_{24} + 4x_9x_{34} - 51x_{10} - 5x_{10}^2 + 2x_{10}x_{11} \\
& + 4x_{10}x_{19} - 8x_{10}x_{23} - 16x_{10}x_{27} - 8x_{10}x_{57} - 2x_{11} \\
& - 2x_{16} + 3x_{19} - 8x_{19}x_{23} - 8x_{19}x_{24} - 8x_{19}x_{27} \\
& + 2x_{20} - x_{23} - 8x_{23}^2 + 16x_{23}x_{24} + 9x_{24} \\
& - 8x_{24}^2 - 16x_{24}x_{27} - x_{25} - 4x_{25}^2 + x_{26} \\
& + 10x_{27} - 4x_{27}^2 + x_{28} + x_{29} + x_{31} \\
& + 4x_{34}^2 - x_{35} - x_{36} - x_{39} + x_{43} \\
& - 8x_{57}^2 + 2x_{58} + 4x_{58}^2 + x_{59} - x_{60} \\
& - 9x_{71} + x_{71}^2 - x_{72} + 2x_{131}^2 + 2x_{131}x_{136} \\
& + 2x_{131}x_{161} + 2x_{131}x_{170} + 2x_{133}^2 - 2x_{134}^2 + 2x_{135} \\
& - 2x_{137} - x_{138} - x_{142}^2 + 2x_{142}x_{144} + x_{143}^2 \\
& - x_{145}^2 + 2x_{145}x_{147} + x_{146}^2 - x_{148}^2 + 2x_{148}x_{150} \\
& + x_{149}^2 - x_{151}^2 + 2x_{151}x_{153} + x_{152}^2 - x_{154}^2 \\
& + 2x_{154}x_{156} + x_{155}^2 - x_{157}^2 + 2x_{157}x_{159} + x_{158}^2 \\
& + 2x_{160} - 2x_{162} - x_{163}^2 + 2x_{163}x_{165} + x_{164}^2 \\
& - x_{166}^2 + 2x_{166}x_{168} + x_{167}^2 + 2x_{169} - 2x_{171} \\
& - x_{172}^2 + 2x_{172}x_{174} + x_{173}^2 - x_{175}^2 + 2x_{175}x_{177} \\
& + x_{176}^2 - 4x_{178}^2 + x_{179}^2 + x_{180}^2 + 2x_{181}^2 \\
& + 2x_{182}^2 - x_{189}^2 - 2x_{190}^2 + x_{207}^2 - x_{208}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{14} = \; & -2 + x_0 - x_1 + x_6x_7 + 2x_8 \\
& - 9x_8^2 + 19x_8x_9 + 5x_8x_{10} - 4x_8x_{23} - 8x_8x_{24} \\
& - 32x_8x_{37} - 13x_9 + 9x_9^2 + 5x_9x_{10} - 8x_9x_{23} \\
& + 4x_9x_{24} + 12x_{10} - 4x_{10}^2 + 2x_{10}x_{11} - 4x_{10}x_{19} \\
& - 8x_{10}x_{23} - 20x_{10}x_{27} - 24x_{10}x_{37} + x_{11}^2 - 8x_{19}x_{23} \\
& + 9x_{23} - 8x_{23}^2 + 16x_{23}x_{24} - x_{24} - 8x_{24}^2 \\
& - 16x_{24}x_{27} - 8x_{24}x_{37} - 8x_{25}^2 + 2x_{26} - 2x_{27} \\
& - 4x_{27}^2 - x_{29} - 4x_{29}^2 + x_{30} + x_{31} \\
& - 36x_{37} - 4x_{37}^2 + x_{38} - x_{39} + x_{42} \\
& - 2x_{43} - x_{59} + 7x_{71} - 4x_{178}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{15} = \; & 2 + x_0 + 2x_1 - x_2 - x_4x_5 \\
& - x_6x_7 + 11x_8 + 4x_8^2 + 16x_8x_9 - 2x_8x_{10} \\
& + 8x_8x_{23} - 8x_8x_{24} - 4x_8x_{39} - 16x_9 - 9x_9^2 \\
& + 6x_9x_{10} - 8x_9x_{24} - 32x_9x_{36} + 12x_{10} - 7x_{10}^2
\end{aligned}
$$

$$
\begin{aligned}
&+ 2x_{10}x_{11} + 12x_{10}x_{19} - 16x_{10}x_{23} + 8x_{10}x_{24} - 4x_{10}x_{27} \\
&- 24x_{10}x_{36} - 8x_{10}x_{39} + x_{11}^2 - 4x_{19} - 8x_{19}x_{23} \\
&+ 16x_{19}x_{24} + 8x_{19}x_{27} + x_{20} + 2x_{23} - 4x_{23}^2 \\
&- 8x_{23}x_{36} - 10x_{24} - 13x_{27} - x_{28} + x_{31} \\
&+ 4x_{36} - 4x_{36}^2 + x_{37} - x_{39} - 4x_{39}^2 \\
&+ x_{40} + x_{42} - x_{43} - x_{59} + 6x_{71} \\
&+ x_{131}^2 + x_{131}x_{136} + x_{131}x_{161} + x_{131}x_{170} + x_{133}^2 \\
&- x_{134}^2 + x_{135} - x_{137} + x_{160} - x_{162} \\
&+ x_{169} - x_{171} - 4x_{178}^2 + x_{179}^2 + x_{180}^2 \\
&+ x_{181}^2 + x_{182}^2 - x_{189}^2 - x_{190}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{16} = \; &- 1 - x_0 - 17x_8 + 6x_8^2 + 27x_8x_9 \\
&- 3x_8x_{10} - 8x_8x_{19} - 60x_8x_{23} - 16x_8x_{24} - 32x_8x_{31} \\
&- 4x_8x_{39} + 6x_9 - 15x_9^2 + 7x_9x_{10} + 32x_9x_{19} \\
&+ 8x_9x_{23} + 17x_{10} - 8x_{10}^2 + 2x_{10}x_{11} - 32x_{10}x_{19} \\
&- 24x_{10}x_{23} + 16x_{10}x_{24} - 20x_{10}x_{27} - 40x_{10}x_{31} - 8x_{10}x_{39} \\
&+ x_{11}^2 + 12x_{19} - 8x_{19}x_{23} - 8x_{19}x_{24} - 4x_{20} \\
&- 5x_{23} - 16x_{23}^2 + 8x_{23}x_{24} - 16x_{23}x_{31} - 8x_{24} \\
&- 4x_{24}^2 - 16x_{24}x_{27} - 8x_{24}x_{31} + x_{25} - 4x_{27} \\
&- 4x_{27}^2 - 2x_{29} - 4x_{29}^2 + x_{30} - 43x_{31} \\
&- 4x_{31}^2 + x_{32} + x_{36} - x_{39} - 4x_{39}^2 \\
&+ x_{40} + x_{42} + x_{43} - 2x_{59} + 13x_{71} \\
&+ 4x_{178}^2
\end{aligned}
$$

$$
R_{17} = \; - 4x_{19}^2 + x_{20}
$$

$$
\begin{aligned}
R_{18} = \; &x_8 - x_8^2 + 8x_8x_{19} - 4x_8x_{20} + 16x_{19}x_{20} \\
&- 4x_{20} - 4x_{20}^2 + x_{33}
\end{aligned}
$$

$$
R_{19} = 1 + 2x_{33} - 4x_{33}^2 + x_{34}
$$

$$
R_{20} = \; - x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35}
$$

$$
\begin{aligned}
R_{21} = \; &2 + 2x_1 - x_2 - x_4x_5 - x_6x_7 \\
&+ 4x_8 + 6x_8^2 + 5x_8x_9 - x_8x_{10} + 16x_8x_{19} \\
&- 4x_9 + 8x_9^2 - 5x_9x_{10} + 32x_9x_{19} - 11x_{10} \\
&- 3x_{10}^2 + 2x_{10}x_{11} + 4x_{10}x_{19} + x_{11}^2 + 25x_{19} \\
&- 3x_{20} - 2x_{23} + x_{36} - x_{39} - x_{59} \\
&+ 7x_{71} + x_{131}^2 + x_{131}x_{136} + x_{133}^2 - x_{134}^2 \\
&+ x_{135} - x_{137}
\end{aligned}
$$

$$
\begin{aligned}
R_{22} = \; &1 + x_0 - 3x_8 - 2x_8^2 - 5x_8x_9 \\
&+ x_8x_{10} + 11x_9 - 15x_9^2 - 7x_9x_{10} - 8x_9x_{23} \\
&- 32x_9x_{36} + 16x_9x_{43} + 4x_9x_{47} + 6x_{10} + 2x_{10}^2 \\
&- 16x_{10}x_{23} - 24x_{10}x_{36} + 4x_{10}x_{43} + 8x_{10}x_{47} + x_{19} \\
&- 27x_{23} + 8x_{23}^2 - 8x_{23}x_{36} + 16x_{23}x_{43} + x_{24} \\
&+ 4x_{36} - 4x_{36}^2 + x_{37} + x_{43} + 4x_{43}^2 \\
&- x_{44} + 4x_{47}^2 - x_{48} + 3x_{71} - 4x_{178}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{23} = \; & -4 - x_0 - 3x_1 + x_2 + 2x_2x_{208} \\
& + x_3^2 - 2x_3x_{131} + x_4x_5 + 2x_6x_7 - 11x_8 \\
& - 7x_8^2 - 2x_8x_9 - 4x_8x_{10} + 24x_8x_{19} - 8x_8x_{23} \\
& + 17x_9 + 8x_9^2 + 8x_9x_{10} + 24x_9x_{23} - 4x_9x_{24} \\
& - 16x_9x_{43} + 4x_{10}^2 - 2x_{10}x_{11} + 32x_{10}x_{19} + 8x_{10}x_{23} \\
& - 16x_{10}x_{24} - 4x_{10}x_{43} + 2x_{11} + 21x_{19} + 8x_{19}x_{24} \\
& + x_{20} + 16x_{23} - 8x_{23}x_{24} - 16x_{23}x_{43} - 6x_{24} \\
& - 4x_{24}^2 - x_{25} - 8x_{25}^2 + 2x_{26} + x_{27} \\
& + x_{29} + 4x_{29}^2 - x_{30} + x_{39} - 7x_{43} \\
& - 4x_{43}^2 + x_{44} - x_{71}^2 + x_{72} - x_{131}^2 \\
& - x_{131}x_{136} - x_{131}x_{161} - 2x_{131}x_{170} - x_{133}^2 + x_{134}^2 \\
& - x_{135} + x_{137} + x_{138} + x_{142}^2 - 2x_{142}x_{144} \\
& - x_{143}^2 + x_{145}^2 - 2x_{145}x_{147} - x_{146}^2 + x_{148}^2 \\
& - 2x_{148}x_{150} - x_{149}^2 + x_{151}^2 - 2x_{151}x_{153} - x_{152}^2 \\
& + x_{154}^2 - 2x_{154}x_{156} - x_{155}^2 + x_{157}^2 - 2x_{157}x_{159} \\
& - x_{158}^2 - x_{160} + x_{162} + x_{163}^2 - 2x_{163}x_{165} \\
& - x_{164}^2 + x_{166}^2 - 2x_{166}x_{168} - x_{167}^2 - 2x_{169} \\
& + 2x_{171} + x_{172}^2 - 2x_{172}x_{174} - x_{173}^2 + x_{175}^2 \\
& - 2x_{175}x_{177} - x_{176}^2 + 4x_{178}^2 - x_{179}^2 - x_{180}^2 \\
& - x_{181}^2 - x_{182}^2 + x_{189}^2 + x_{190}^2 - x_{207}^2 \\
& + x_{208}^2
\end{aligned}
$$

$$
R_{24} = \; - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39}
$$

$$
\begin{aligned}
R_{25} = \; & -1 - x_1 + x_6x_7 - 2x_8 + x_8x_9 \\
& + x_8x_{10} - 8x_8x_{24} - 4x_8x_{39} + x_9^2 + x_9x_{10} \\
& - 8x_9x_{24} + x_{10}^2 - 8x_{10}x_{39} - x_{23} - 8x_{23}x_{24} \\
& + 5x_{24} - 4x_{24}^2 + 4x_{25}^2 - x_{26} - 4x_{39}^2 \\
& + x_{40} + x_{43} - x_{71}
\end{aligned}
$$

$$
\begin{aligned}
R_{26} = \; & -2x_8 - 2x_8^2 + 3x_8x_9 - 11x_8x_{10} + 8x_8x_{23} \\
& - 8x_8x_{24} - 28x_8x_{39} - 16x_8x_{40} + 4x_8x_{52} + 4x_9 \\
& - x_9^2 + 3x_9x_{10} + 24x_9x_{24} - 5x_{10} - 5x_{10}^2 \\
& - 16x_{10}x_{23} - 8x_{10}x_{24} + 20x_{10}x_{27} - 24x_{10}x_{39} - 16x_{10}x_{40} \\
& + 8x_{10}x_{52} - x_{19} + 2x_{23} - 4x_{23}^2 + 16x_{23}x_{24} \\
& - 27x_{24} + 8x_{24}^2 + 16x_{24}x_{27} - 32x_{24}x_{39} - 16x_{24}x_{40} \\
& - 8x_{25}^2 + 2x_{26} + x_{27} + 4x_{27}^2 - 4x_{39} \\
& - 12x_{39}^2 - 16x_{39}x_{40} - 3x_{40} - 4x_{40}^2 + x_{41} \\
& - x_{42} - x_{43} + 4x_{52}^2 - x_{53} + 7x_{71}
\end{aligned}
$$

$$
\begin{aligned}
R_{27} = \; & 2 + x_1 - x_6x_7 + 2x_8^2 + 2x_8x_9 \\
& + 2x_8x_{10} + x_9^2 + x_{10} - 3x_{10}^2 + 2x_{10}x_{11} \\
& + 16x_{10}x_{23} - 24x_{10}x_{24} - 20x_{10}x_{27} + x_{11}^2 - x_{23} \\
& + 4x_{23}^2 + 8x_{23}x_{24} - 2x_{24} - 12x_{24}^2 - 16x_{24}x_{27} \\
& - x_{27} - 4x_{27}^2 - x_{39} + x_{42} - 2x_{71} \\
& + x_{131}^2 + x_{133}^2 - x_{134}^2
\end{aligned}
$$

$$\begin{aligned} R_{28} = & -1 - 3x_8 - x_8^2 - x_8x_9 - x_8x_{10} \\ & + 4x_9 - 3x_9^2 + 3x_9x_{10} - 8x_9x_{23} + 3x_{10}^2 \\ & + x_{19} + 6x_{23} - 4x_{23}^2 + x_{43} - x_{71} \end{aligned}$$

$$\begin{aligned} R_{29} = & 2 + 2x_1 - x_2 - x_4x_5 - x_6x_7 \\ & + 6x_8 + 2x_8^2 + 2x_8x_9 + 2x_8x_{10} - 8x_9 \\ & - 6x_9^2 - 6x_9x_{10} - 16x_9x_{23} - 16x_9x_{43} - 3x_{10}^2 \\ & - 8x_{10}x_{23} - 4x_{10}x_{43} - 2x_{19} - 16x_{23} - 8x_{23}^2 \\ & - 16x_{23}x_{43} - 2x_{43} - 4x_{43}^2 + x_{44} + 2x_{71} \\ & + x_{131}^2 + x_{131}x_{136} + x_{131}x_{161} + x_{131}x_{170} + x_{133}^2 \\ & - x_{134}^2 + x_{135} - x_{137} + x_{160} - x_{162} \\ & + x_{169} - x_{171} + x_{179}^2 + x_{180}^2 - x_{189}^2 \end{aligned}$$

$$\begin{aligned} R_{30} = & 1 + x_1 - x_6x_7 + x_8^2 + 2x_8x_{10} \\ & + x_9 - 2x_9x_{10} - 4x_9x_{44} - 2x_{10}^2 - 8x_{10}x_{44} \\ & - x_{23} - x_{24} - 4x_{44}^2 + x_{45} \end{aligned}$$

$$R_{31} = -1 - 4x_{45} - 4x_{45}^2 + x_{46}$$

$$\begin{aligned} R_{32} = & 1 - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} + 8x_{45} \\ & - 8x_{45}^2 - 16x_{45}x_{46} - 2x_{46} - 4x_{46}^2 + x_{47} \end{aligned}$$

$$\begin{aligned} R_{33} = & 2 + x_1 - x_6x_7 + x_8^2 + 2x_8x_{10} \\ & + x_9 - 2x_9x_{10} - 4x_9x_{47} - 2x_{10}^2 - 8x_{10}x_{47} \\ & - x_{23} - x_{24} - 4x_{47}^2 + x_{48} \end{aligned}$$

$$R_{34} = 1 - 4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49}$$

$$\begin{aligned} R_{35} = & x_8 - x_8^2 - 8x_8x_{42} - 4x_8x_{49} - 12x_{42}^2 \\ & + 8x_{42}x_{48} - 16x_{42}x_{49} + 4x_{48}^2 - x_{49} - 4x_{49}^2 \\ & + x_{50} \end{aligned}$$

$$R_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$\begin{aligned} R_{37} = & 1 - x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} + 8x_{50} \\ & - 8x_{50}^2 - 16x_{50}x_{51} - 2x_{51} - 4x_{51}^2 + x_{52} \end{aligned}$$

$$\begin{aligned} R_{38} = & x_8 + x_8^2 - 4x_8x_{52} - 2x_{10}^2 - 8x_{10}x_{52} \\ & - 2x_{24} - 4x_{52}^2 + x_{53} \end{aligned}$$

$$\begin{aligned} R_{39} = & -2 - 2x_1 + x_2 + x_4x_5 + x_6x_7 \\ & - x_8^2 - 2x_8x_{10} + x_9^2 + 2x_9x_{10} - x_{23} \\ & + x_{24} - 12x_{53} - 4x_{53}^2 + x_{54} - x_{131}^2 \\ & - x_{131}x_{136} - x_{131}x_{161} - x_{133}^2 + x_{134}^2 - x_{135} \\ & + x_{137} - x_{160} + x_{162} \end{aligned}$$

$$\begin{aligned}R_{40} = & -4x_9 - x_9^2 - 2x_9x_{10} - 16x_9x_{53} - 8x_9x_{54} \\ & - x_{10}^2 - 3x_{23} - 16x_{23}x_{53} - 8x_{23}x_{54} - x_{43} \\ & + 8x_{53} - 16x_{53}^2 - 16x_{53}x_{54} + 4x_{54} - 4x_{54}^2 \\ & + x_{55}\end{aligned}$$

$$\begin{aligned}R_{41} = & -2 - 2x_1 + x_2 + x_4x_5 + x_6x_7 \\ & + 6x_8 + 2x_8^2 + 4x_8x_9 + x_9 - 5x_9^2 \\ & + 4x_9x_{10} + 16x_9x_{43} - 20x_9x_{55} - x_{10}^2 + 8x_{10}x_{23} \\ & + 4x_{10}x_{43} - 2x_{19} - 6x_{23} + 4x_{23}^2 + 16x_{23}x_{43} \\ & - 16x_{23}x_{55} - x_{43} + 4x_{43}^2 - x_{44} - 12x_{48} \\ & - 4x_{48}^2 - 4x_{55}^2 + x_{56}\end{aligned}$$

$$\begin{aligned}R_{42} = & 3 + 2x_1 - x_2 - x_4x_5 - x_6x_7 \\ & - 3x_8 - 2x_8x_9 + 2x_8x_{10} - 4x_9 - 2x_9^2 \\ & - 2x_9x_{10} + x_{19} + x_{23} - x_{24} + 24x_{48} \\ & - 8x_{48}^2 - 16x_{48}x_{56} - 2x_{56} - 4x_{56}^2 + x_{57} \\ & + x_{131}^2 + x_{131}x_{136} + x_{131}x_{161} + x_{131}x_{170} + x_{133}^2 \\ & - x_{134}^2 + x_{135} - x_{137} - x_{154}^2 + 2x_{154}x_{156} \\ & + x_{155}^2 - x_{157}^2 + 2x_{157}x_{159} + x_{158}^2 + x_{160} \\ & - x_{162} - x_{163}^2 + 2x_{163}x_{165} + x_{164}^2 + x_{169} \\ & - x_{171} + x_{179}^2 + x_{180}^2 + x_{181}^2 + x_{182}^2 \\ & - x_{189}^2 - x_{190}^2\end{aligned}$$

$$\begin{aligned}R_{43} = & -2 - x_1 + x_6x_7 + x_8 - x_8^2 \\ & - x_8x_9 - x_8x_{10} + 4x_9 + x_9x_{10} + x_{10} \\ & + x_{10}^2 - 4x_{10}x_{57} - x_{19} - x_{23} - 4x_{57}^2 \\ & + x_{58} + 12x_{59} - 4x_{59}^2 - x_{71}\end{aligned}$$

$$\begin{aligned}R_{44} = & 2 + x_0 + x_1 - x_6x_7 + 15x_8 \\ & + 6x_8^2 - 6x_8x_9 - 2x_8x_{10} + 16x_8x_{19} + 8x_8x_{23} \\ & - 38x_9^2 + 8x_9x_{10} + 32x_9x_{19} + 8x_9x_{23} + 4x_9x_{34} \\ & + 11x_{10} + 5x_{10}^2 - 2x_{10}x_{11} - 4x_{10}x_{19} - x_{11}^2 \\ & - 2x_{19} - x_{20} - 8x_{23} + 4x_{23}^2 + x_{24} \\ & + 4x_{25}^2 - x_{26} + 4x_{34}^2 - x_{35} + x_{39} \\ & + x_{43} + x_{59} - 8x_{71} + x_{131}^2 + x_{133}^2 \\ & - x_{134}^2 - 4x_{178}^2\end{aligned}$$

$$\begin{aligned}R_{45} = & -1 - x_1 + x_6x_7 - 2x_{10} + 2x_{10}^2 \\ & + 8x_{10}x_{57} + 8x_{57}^2 - 2x_{58} - 4x_{58}^2 + x_{60}\end{aligned}$$

$$R_{46} = -1 - 4x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{47} = -x_2x_{13} + x_{62}$$

$$R_{48} = x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}$$

$$R_{49} = x_{12} - x_{13} - 2x_{26}x_{62} - x_{64}$$

$$R_{50} = -x_2x_{13} + x_{12} - x_{13} - 2x_{28}x_{62} - x_{62} - x_{65}$$

$$R_{51} = x_{12} - x_{13} - 2x_{30}x_{62} - x_{66}$$

$$R_{52} = x_2x_{13} + x_{12} - x_{13} - 2x_{32}x_{62} + x_{62} - x_{67}$$

$$R_{53} = x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68}$$

$$R_{54} = x_{12} - x_{13} - 2x_{38}x_{62} - x_{69}$$

$$R_{55} = -x_2x_{13} + 2x_{28}x_{62} - 2x_{41}x_{62} - x_{62} + x_{65} - x_{70}$$

$$R_{56} = -x_8x_9 + x_8x_{10} + x_9x_{10} - x_{10}^2 + x_{71}$$

$$R_{57} = 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72}$$

$$R_{58} = -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 + x_{73} - x_{74}^2$$

$$R_{59} = -4x_{60}x_{62} - 2x_{61}x_{62} + x_{74}$$

$$R_{60} = -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77}$$

$$R_{61} = -2 + 4x_2 + 4x_2x_8 + 8x_2x_9 - 2x_2x_{19} + x_3x_{18} - 2x_4x_5 + 2x_6x_7 - 2x_{16} + x_{75} + 2x_{209}x_{211} + x_{211}$$

$$R_{62} = -x_8x_{201} - x_{12} + x_{76}$$

$$R_{63} = x_2x_{13} + x_9x_{201} + x_{12} - x_{13} - 2x_{19}x_{62} + x_{62} + x_{78}$$

$$R_{64} = -x_{12}x_{63} + x_{13}x_{63} + x_{79}$$

$$R_{65} = -x_{64}x_{79} + x_{80}$$

$$R_{66} = -x_{65}x_{80} + x_{81}$$

$$R_{67} = -x_{66}x_{81} + x_{82}$$

$$R_{68} = -x_{67}x_{82} + x_{83}$$

$$R_{69} = -x_{68}x_{83} + x_{84}$$

$$R_{70} = -x_{69}x_{84} + x_{85}$$

$$R_{71} = -x_{70}x_{85} + x_{86}$$

$$R_{72} = -x_{77}x_{86} + x_{87}$$

$$R_{73} = -x_{78}x_{87} + x_{88}$$

$$R_{74} = -x_{88}^2 + x_{89}$$

$$\begin{aligned} R_{75} = & -1 - x_2 + x_3x_{18} - x_4 - 2x_4x_{16} \\ & + x_{90} \end{aligned}$$

$$R_{76} = -x_{14}x_{90} + x_{91}$$

$$\begin{aligned} R_{77} = & -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} \\ & - x_{12}^2 - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{93} \\ & - x_{73}x_{89} - x_{91}^2 - x_{92}^2 \end{aligned}$$

$$R_{78} = -x_4x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$

$$R_{79} = -x_{94}x_{97} + x_{97} + x_{99}$$

$$R_{80} = -x_3x_{120} + x_{95}x_{193} - x_{98} - x_{99}x_{120}$$

$$R_{81} = x_{100}x_{193} - x_{101}x_{123} - x_{102}$$

$$R_{82} = x_{103}x_{193} - x_{104}x_{123} - x_{105}$$

$$R_{83} = x_{106}x_{193} - x_{107}x_{123} - x_{108}$$

$$R_{84} = x_{109}x_{193} - x_{110}x_{123} - x_{111}$$

$$R_{85} = x_{112}x_{194} - x_{114}x_{123} - x_{115}$$

$$R_{86} = x_{116}x_{194} - x_{117}x_{123} - x_{118}$$

$$R_{87} = -x_{17}x_{120} + x_{123}$$

$$R_{88} = x_{119} - x_{121}x_{123} - x_{122}$$

$$R_{89} = -x_{123}x_{125} + x_{124} - x_{126}$$

$$R_{90} = -x_{123}x_{128} + x_{127}x_{193} - x_{129}$$

$$R_{91} = x_{130} - 2x_{130}x_{132} + x_{133}$$

$$R_{92} = -x_{130}x_{133} - x_{133} + x_{134}$$

$$R_{93} = -1 + x_{131}^2 + x_{133}^2 - x_{134}^2$$

$$R_{94} = 1 - x_{131}x_{136} - x_{135} + x_{137}$$

$$\begin{aligned} R_{95} = & -x_{131}x_{136} + x_{135} - x_{135}^2 + 2x_{135}x_{137} + x_{136}^2 \\ & + x_{137} \end{aligned}$$

$$R_{96} = 1 + x_3^2 - 2x_3x_{131} + x_{138}$$

$$R_{97} = -x_3x_{170} + x_{17} + x_{138}x_{139} - x_{169}$$

$$\begin{aligned} R_{98} = & 2 - 2x_2 - 4x_2^2 - 16x_2x_{16} + 4x_2x_{131} \\ & - 5x_3x_{18} - x_4 + x_4x_5 - 2x_4x_{16} - x_6x_7 \\ & - 4x_{16} - 8x_{16}^2 + 8x_{16}x_{131} + x_{90} + 3x_{131} \\ & + 2x_{136}^2 - x_{140} + 2x_{143}^2 + 2x_{146}^2 + 2x_{149}^2 \\ & + 2x_{152}^2 + 2x_{155}^2 + 2x_{158}^2 + 2x_{161}^2 + 2x_{164}^2 \\ & + 2x_{167}^2 + 2x_{170}^2 + 2x_{173}^2 - x_{176}^2 - x_{179}x_{186} \\ & + x_{183} - x_{191} + 2x_{195}^2 + 2x_{196}^2 + 2x_{197} \\ & + 2x_{197}^2 \end{aligned}$$

$$\begin{aligned} R_{99} = & -1 - 2x_2x_{170} - 4x_{16}x_{170} + x_{94} + x_{140}x_{141} \\ & - x_{169} - 2x_{170} \end{aligned}$$

$$R_{100} = -x_{131}x_{143} - x_{142} + x_{144}$$

$$R_{101} = -1 - x_{142}^2 + 2x_{142}x_{144} + x_{143}^2$$

$$R_{102} = -x_{131}x_{146} - x_{145} + x_{147}$$

$$R_{103} = -1 - x_{145}^2 + 2x_{145}x_{147} + x_{146}^2$$

$$R_{104} = -x_{131}x_{149} - x_{148} + x_{150}$$

$$R_{105} = -1 - x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$R_{106} = -x_{131}x_{152} - x_{151} + x_{153}$$

$$R_{107} = -1 - x_{151}^2 + 2x_{151}x_{153} + x_{152}^2$$

$$R_{108} = -x_{131}x_{155} - x_{154} + x_{156}$$

$$R_{109} = -1 - x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$

$$R_{110} = -x_{131}x_{158} - x_{157} + x_{159}$$

$$R_{111} = -1 - x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$

$$R_{112} = 1 - x_{131}x_{161} - x_{160} + x_{162}$$

$$\begin{aligned} R_{113} = & -x_{131}x_{161} + x_{160} - x_{160}^2 + 2x_{160}x_{162} + x_{161}^2 \\ & + x_{162} \end{aligned}$$

$$R_{114} = -x_{131}x_{164} - x_{163} + x_{165}$$

$$R_{115} = -1 - x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$

$$R_{116} = -x_{131}x_{167} - x_{166} + x_{168}$$

$$R_{117} = -1 - x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$

$$R_{118} = 1 - x_{131}x_{170} - x_{169} + x_{171}$$

$$\begin{aligned} R_{119} = & -x_{131}x_{170} + x_{169} - x_{169}^2 + 2x_{169}x_{171} + x_{170}^2 \\ & + x_{171} \end{aligned}$$

$$R_{120} = -x_{131}x_{173} - x_{172} + x_{174}$$

$$R_{121} = -1 - x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$

$$R_{122} = -x_{131}x_{176} - x_{175} + x_{177}$$

$$R_{123} = -1 - x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$

$$\begin{aligned} R_{124} = & -1 - 2x_0 - 2x_0^2 - 2x_0x_2 - x_1^2 \\ & - 2x_2^2 - 4x_2x_{131} - x_3^2 - x_4^2 - x_8^2 \\ & + 2x_9x_{10} - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\ & - x_{15}^2 - 4x_{16}^2 - 8x_{16}x_{131} - x_{17}^2 - x_{23} \\ & + 2x_{94} - 2x_{94}^2 - x_{95}^2 - x_{96}^2 - x_{100}^2 \\ & - x_{103}^2 - x_{106}^2 - x_{109}^2 - x_{112}^2 - x_{113}^2 \\ & - x_{116}^2 - x_{119}^2 - x_{120}^2 - x_{124}^2 - x_{127}^2 \\ & + x_{130} - 3x_{131} + x_{140} + x_{179}x_{186} - x_{183} \\ & - 2x_{210} \end{aligned}$$

$$\begin{aligned} R_{125} = & -8 + x_0 + x_0^2 + 2x_0x_2 - 2x_0x_{136} \\ & - 2x_0x_{143} - 4x_1 + x_1^2 - 2x_1x_{170} + 3x_2 \\ & + x_2^2 - 2x_2x_{143} + 2x_2x_{206} + 2x_2x_{208} + x_3^2 \\ & - 2x_3x_{18} - 2x_3x_{131} + x_3x_{170} + x_4^2 + x_4x_5 \\ & + 2x_4x_{16} - 2x_4x_{161} + 3x_6x_7 + 3x_8 + x_8x_9 \\ & - 3x_8x_{10} - 2x_8x_{146} + 4x_9 + 4x_9^2 + 5x_9x_{10} \\ & - 2x_9x_{149} - 2x_{10}x_{152} + 2x_{11} + 2x_{11}^2 - 2x_{11}x_{155} \\ & + x_{12}^2 - 2x_{12}x_{158} + x_{13}^2 - 2x_{13}x_{164} + x_{14}^2 \\ & - 2x_{14}x_{167} + x_{15}^2 - 2x_{15}x_{173} + 2x_{16} + 4x_{16}^2 \\ & - 4x_{16}x_{176} - x_{17} - x_{19} - 2x_{23} + 2x_{24} \\ & + x_{71} - x_{71}^2 + x_{72} - x_{90} - 3x_{131}^2 \\ & - 3x_{131}x_{136} - x_{131}x_{143} - x_{131}x_{146} - x_{131}x_{149} - x_{131}x_{152} \\ & - x_{131}x_{155} - x_{131}x_{158} - 3x_{131}x_{161} - x_{131}x_{164} - x_{131}x_{167} \\ & - 2x_{131}x_{170} - x_{131}x_{173} - x_{131}x_{176} - x_{131}x_{192} - 3x_{133}^2 \\ & + 3x_{134}^2 - 3x_{135} + x_{136}^2 + 3x_{137} + x_{137}^2 \\ & + x_{138} - x_{138}x_{139} - x_{142} + x_{142}^2 - 2x_{142}x_{144} \\ & - x_{144} + x_{144}^2 - x_{145} + x_{145}^2 - 2x_{145}x_{147} \\ & - x_{147} + x_{147}^2 - x_{148} + x_{148}^2 - 2x_{148}x_{150} \\ & - x_{150} + x_{150}^2 - x_{151} + x_{151}^2 - 2x_{151}x_{153} \end{aligned}$$

$$
\begin{aligned}
&- x_{153} + x_{153}^2 - x_{154} + x_{154}^2 - 2x_{154}x_{156} \\
&- x_{156} + x_{156}^2 - x_{157} + x_{157}^2 - 2x_{157}x_{159} \\
&- x_{159} + x_{159}^2 - 3x_{160} + x_{161}^2 + 3x_{162} \\
&+ x_{162}^2 - x_{163} + x_{163}^2 - 2x_{163}x_{165} - x_{165} \\
&+ x_{165}^2 - x_{166} + x_{166}^2 - 2x_{166}x_{168} - x_{168} \\
&+ x_{168}^2 - 3x_{169} + x_{169}^2 - 2x_{169}x_{171} + 2x_{171} \\
&+ x_{171}^2 - x_{172} + x_{172}^2 - 2x_{172}x_{174} - x_{174} \\
&+ x_{174}^2 - x_{175} + x_{175}^2 - 2x_{175}x_{177} - x_{177} \\
&+ x_{177}^2 - 4x_{178}^2 - 2x_{179}^2 - 2x_{180}^2 - 2x_{181}^2 \\
&- 2x_{182}^2 + 2x_{189}^2 + 2x_{190}^2 + x_{192} - x_{207}^2 \\
&+ x_{208}^2 + 2x_{209}x_{211} + x_{210} + x_{211}
\end{aligned}
$$

$$R_{126} = \ - x_{131}x_{180} + x_{189}$$

$$R_{127} = \ - 1 + x_{179}^2 + x_{180}^2 - x_{189}^2$$

$$R_{128} = \ - x_{182}x_{183} + x_{190}$$

$$R_{129} = \ - 1 + x_{181}^2 + x_{182}^2 - x_{190}^2$$

$$R_{130} = \ - x_{176}^2 + x_{191}$$

$$R_{131} = x_{180} - 2x_{185}x_{191} - x_{191}$$

$$
\begin{aligned}
R_{132} = \ &- 1 - 3x_{180} - 8x_{180}x_{184} + x_{183} - 2x_{185}x_{191} \\
&- x_{191}
\end{aligned}
$$

$$R_{133} = \ - x_{131} - x_{179}x_{186} + x_{183}$$

$$R_{134} = \ - x_{177} - x_{179}x_{187} + x_{181}$$

$$R_{135} = 1 + x_2 - x_3x_{18} - 4x_{176}x_{188} + x_{182}$$

$$
\begin{aligned}
R_{136} = \ &2 + x_0^2 + 2x_0x_2 - 2x_0x_{136} - 2x_0x_{143} \\
&+ 2x_1 + x_1^2 - 2x_1x_{170} - x_2 + x_2^2 \\
&- 2x_2x_{143} + x_4^2 - x_4x_5 - 2x_4x_{161} - x_6x_7 \\
&+ x_8^2 - 2x_8x_{146} - 2x_9x_{10} - 2x_9x_{149} - 2x_{10}x_{152} \\
&+ x_{11}^2 - 2x_{11}x_{155} + x_{12}^2 - 2x_{12}x_{158} + x_{13}^2 \\
&- 2x_{13}x_{164} + x_{14}^2 - 2x_{14}x_{167} + x_{15}^2 - 2x_{15}x_{173} \\
&- 4x_{16}x_{176} + x_{23} + x_{131}^2 + x_{131}x_{136} - x_{131}x_{143} \\
&- x_{131}x_{146} - x_{131}x_{149} - x_{131}x_{152} - x_{131}x_{155} - x_{131}x_{158} \\
&+ x_{131}x_{161} - x_{131}x_{164} - x_{131}x_{167} - x_{131}x_{173} - x_{131}x_{176} \\
&- x_{131}x_{192} + x_{133}^2 - x_{134}^2 + x_{135} - x_{137} \\
&+ x_{137}^2 - x_{142} - x_{144} + x_{144}^2 - x_{145} \\
&- x_{147} + x_{147}^2 - x_{148} - x_{150} + x_{150}^2 \\
&- x_{151} - x_{153} + x_{153}^2 - x_{154} - x_{156} \\
&+ x_{156}^2 - x_{157} - x_{159} + x_{159}^2 + x_{160} \\
&- x_{162} + x_{162}^2 - x_{163} - x_{165} + x_{165}^2 \\
&- x_{166} - x_{168} + x_{168}^2 + x_{171}^2 - x_{172} \\
&- x_{174} + x_{174}^2 - x_{175} + x_{176}^2 - x_{177} \\
&+ x_{177}^2 + x_{191} + x_{192} - x_{195}^2 - x_{196}^2 \\
&- x_{197} - x_{197}^2 + x_{210}
\end{aligned}
$$

$$R_{137} = -x_{96}x_{193} + x_{120}$$

$$R_{138} = -x_{113}x_{194} + x_{120}$$

$$\begin{aligned} R_{139} = & -1 - 4x_{16}^2 - x_{136}^2 - x_{143}^2 - x_{146}^2 \\ & - x_{149}^2 - x_{152}^2 - x_{155}^2 - x_{158}^2 - x_{161}^2 \\ & - x_{164}^2 - x_{167}^2 - x_{170}^2 - x_{173}^2 + x_{176}^2 \\ & - x_{195}^2 - x_{196}^2 - x_{197} - x_{197}^2 \end{aligned}$$

$$\begin{aligned} R_{140} = & -2 - x_1 + x_6x_7 - 4x_{98}^2 - 4x_{102}^2 \\ & - 4x_{105}^2 - 4x_{108}^2 - 4x_{111}^2 - 4x_{115}^2 - 4x_{118}^2 \\ & + x_{120}^2 - 4x_{122}^2 - 4x_{126}^2 - 4x_{129}^2 - x_{198}^2 \\ & - x_{199}^2 + 3x_{200} - x_{200}^2 \end{aligned}$$

$$R_{141} = -x_2x_{12} + x_{201}$$

$$\begin{aligned} R_{142} = & x_{94}x_{203} - 2x_{95}x_{143} - 2x_{100}x_{146} - 2x_{103}x_{149} - 2x_{106}x_{152} \\ & - 2x_{109}x_{155} - 2x_{112}x_{158} - 2x_{116}x_{164} - 2x_{119}x_{167} - 2x_{124}x_{173} \\ & - 2x_{127}x_{176} - 2x_{131}x_{203} + 2x_{143}x_{144} + 2x_{146}x_{147} + 2x_{149}x_{150} \\ & + 2x_{152}x_{153} + 2x_{155}x_{156} + 2x_{158}x_{159} + 2x_{164}x_{165} + 2x_{167}x_{168} \\ & + 2x_{173}x_{174} + 2x_{176}x_{177} + x_{202} \end{aligned}$$

$$\begin{aligned} R_{143} = & -x_{94}x_{202} + x_{95}^2 - 2x_{95}x_{142} + x_{100}^2 - 2x_{100}x_{145} \\ & + x_{103}^2 - 2x_{103}x_{148} + x_{106}^2 - 2x_{106}x_{151} + x_{109}^2 \\ & - 2x_{109}x_{154} + x_{112}^2 - 2x_{112}x_{157} + x_{116}^2 - 2x_{116}x_{163} \\ & + x_{119}^2 - 2x_{119}x_{166} + x_{124}^2 - 2x_{124}x_{172} + x_{127}^2 \\ & - 2x_{127}x_{175} + x_{142}^2 - x_{143}^2 + x_{145}^2 - x_{146}^2 \\ & + x_{148}^2 - x_{149}^2 + x_{151}^2 - x_{152}^2 + x_{154}^2 \\ & - x_{155}^2 + x_{157}^2 - x_{158}^2 + x_{163}^2 - x_{164}^2 \\ & + x_{166}^2 - x_{167}^2 + x_{172}^2 - x_{173}^2 + x_{175}^2 \\ & - x_{176}^2 + x_{203} \end{aligned}$$

$$\begin{aligned} R_{144} = & x_{94}x_{205} - 2x_{96}x_{136} - 2x_{113}x_{161} - 2x_{120}x_{170} - 2x_{131}x_{205} \\ & + 2x_{136} + 2x_{136}x_{137} + 2x_{161} + 2x_{161}x_{162} + 2x_{170} \\ & + 2x_{170}x_{171} + x_{204} - x_{205} \end{aligned}$$

$$\begin{aligned} R_{145} = & -x_{94}x_{204} + x_{96}^2 - 2x_{96}x_{135} + x_{113}^2 - 2x_{113}x_{160} \\ & + x_{120}^2 - 2x_{120}x_{169} + x_{135}^2 - x_{136}^2 + x_{160}^2 \\ & - x_{161}^2 + x_{169}^2 - x_{170}^2 + x_{204} + x_{205} \end{aligned}$$

$$R_{146} = -1 - x_2 - 2x_2x_{206} + x_4$$

$$R_{147} = -x_2x_{211} + x_{208}$$

$$R_{148} = -1 - 2x_2x_{208} + x_{207}^2 - x_{208}^2$$

$$R_{149} = x_2 - 2x_{209}x_{211} - x_{211}$$

$$R_{150} = -x_0^2 + x_{210}$$

$$R_{151} = -x_{210}^2 + x_{211}$$